

Postdural puncture headache after intentional or unintentional dural punctures: time to think about risk reduction and acknowledge the burden of sequelae

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Postdural puncture headache (PDPH) is an iatrogenic complication following either an intentional dural puncture (DP) or an unintentional DP during an epidural placement attempt. The exact number of DP, intended or accidental, performed annually worldwide is unknown. However, intentional or unintentional DP affects a broad and diverse population of individuals undergoing diagnostic, therapeutic, or analgesic procedures. A significant number will suffer PDPH,¹ and more serious long-term consequences are increasingly being identified and reported,^{2,3} although the global health burden of PDPH is not yet well described.

Under the American Society of Regional Anesthesia and Pain Medicine (ASRA Pain Medicine) leadership, an international multisociety panel from six societies produced practice guidelines on PDPH after intentional and unintentional DPs. This panel included 21 experts representing ASRA Pain Medicine, the European Society of Regional Anesthesia (ESRA), the Society for Obstetric Anesthesia and Perinatology (SOAP), the Obstetric Anesthetists' Association (OAA), the American Society of Neuroradiology (ASN), and the American Interventional Headache Society (AIHS). This extensive work resulted in a narrative literature review, recommendations for better recognition and follow-up of PDPH, emphasizing informed consent and education of patients and treating teams.^{4,5}

A thorough eight-step process was applied that included (1) selecting the panel, (2) defining the problem (PDPH), (3) conducting a literature review of many publications that generated a guidelines reference list of 460 publications, (4) applying the US Preventative Services Task Force to grade the evidence, (5) narrative writing, (6 and 7) two rounds of modified Delphi approach to reach an agreement on all statements and recommendations, (8) summarizing the findings (reported in a figure).⁵

The problem (PDPH) was defined with 10 questions of clinical relevance, and grading of the evidence resulted in 50 recommendations; 20 relate to epidural blood patch (EBP). The PICO (patient, intervention, comparison, outcome) framework, frequently used as a tool for structuring clinical research questions, was used to establish recommendations and to present levels of evidence. However, within the 10 key questions, multiple subquestions did not necessarily result in a statement or recommendation. A 90% or more agreement was achieved

for most statements and recommendations, with only a few yielding an 80%–89% consensus (agreement is displayed).⁵

WHO ARE THESE PRACTICE GUIDELINES FOR?

The guidelines are intended for all clinicians performing procedures that intentionally or accidentally penetrate the dura and sub-arachnoid. Anesthesiologists, including those providing obstetric anesthesia care, neurologists, neuroradiologists, headache specialists, chronic pain clinicians, emergency medicine clinicians, oncologists, and pediatricians, are likely to be interested. The guidelines do not present recommendations for pediatric patients.

Under question 1, the panel addresses when PDPH should be suspected. This underscores the relevance of these guidelines to any medical practitioner who may encounter a patient with a headache that may be related to an intended or accidental DP, as this problem is not specific to clinicians who perform neuraxial procedures.

In response to each key question, the guidelines recognize that both intentional and unintentional DP cause PDPH. However, they do not clearly and consistently distinguish between these two clinical scenarios and etiologies. Since patient characteristics and procedural risk factors differ, it would be useful to consider intentional and unintentional DPs separately within each key question reviewed. The timeline of presentation (ambulatory vs inpatient procedures) and care pathways may differ too. For example, in the case of intentional DP performed as an outpatient procedure, identification of subsequent PDPH may require more proactive screening than unintentional DP occurring in inpatient obstetric patients.

WHAT STRATEGIES CAN PROMOTE A RISK REDUCTION OF PDPH?

The guidelines emphasize risk reduction of PDPH through a review of patient and obstetric factors (question 2). Thus, specific patients at higher risk of PDPH such as younger female patients, patients with pre-existing headaches, history of headaches, and prior PDPH should be the focus of strategies to minimize the likelihood of PDPH occurring after intentional DP, and of inadvertent DP occurring after epidural placement.



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Two obvious opportunities exist to modify practice among the 11 clinical practice characteristics reviewed. Though not explicitly stated, the first is specific to intentional DP and is related to the use of cutting needles.

Why are cutting needles still in use, given the high level of certainty that “compared with cutting needles, non-cutting spinal needles are associated with decreased risk of PDPH”?⁵ The panel offers a balanced discussion explaining that cutting needles may be easier for the clinician to use, enabling fewer attempts while being less costly. This should be the case in people at risk for PDPH, such as obstetric patients, though might be more difficult to achieve in low-resource settings or when diagnostic procedures necessitate aspiration of large volumes of cerebrospinal fluid (CSF). Cost and availability may preclude a moratorium on the use of cutting needles for intentional DP. However, while the purchase cost of cutting needles is lower, PDPH costs following the use of cutting needles may not be factored into the equation if PDPH is not routinely sought after intentional DP procedures. According to a systematic review that considered three cost-effectiveness studies in neurology patients, the cost of managing PDPH likely outweighs the purchase of cheaper cutting needles.⁶ Another reason clinicians may favor cutting needles is the easier performance of intentional DP, especially if the needle is larger, thus fewer attempts are required. However, it is preferable to use a non-cutting larger needle than a smaller cutting one if the choice is available. Surprisingly, only 85% of the guidelines group agreed that “compared with cutting needles, non-cutting spinal needles are associated with lower PDPH risk”, although all did agree in the first round that “non-cutting needles should be routinely used for all patients undergoing lumbar puncture”. Thus, if a major modifiable global cause of PDPH in intentional DP is needle type, manufacturers, educators, and clinicians must make a sweeping change. This will only happen if PDPH is recognized as an unacceptable outcome with considerable clinical sequelae.

The second opportunity to modify practice relates to the number of attempts and is relevant to both intentional and unintentional DPs. As reported in the guidelines, it makes sense that more attempts translate to a higher rate of PDPH. There are many ways to decrease the number of attempts, including training and supervision and clear “call-for-help rules when a DP or neuraxial block placement is difficult (particularly in training programs where trainees provide unsupervised care).

One should consider all modifiable factors, including the proceduralists’ experience, particularly if performing a procedure on a high-risk patient with needles/techniques associated with a higher risk for PDPH. Should an inexperienced person perform a spinal procedure with a cutting-edge needle on an obstetric (ie, young) patient? Preferably not.

WHAT PROPHYLACTIC MEASURES MAY BE USED TO PREVENT PDPH?

There are many proposed strategies to minimize the development of PDPH after an unintentional DP (question 4); yet unfortunately, there is little evidence to routinely recommend prophylactic measures. Emerging evidence may change this recommendation, but, for example, evidence to support the placement of an intrathecal catheter to reduce the risk of PDPH or prophylactic EBP via indwelling epidural catheter was deemed low. This underscores the importance of proactive management to identify patients with PDPH, since they may go home with recommended prophylactic measures that in practice are unlikely to affect whether PDPH develops, and patients should know this.

WHAT CONSERVATIVE AND PROCEDURAL MEASURES MAY BE USED TO TREAT PDPH?

The evidence does not support treatment of PDPH through the routine use of commonly suggested conservative (question 5) or procedural (question 6) measures. The recommended strategies include hydration (oral or intravenous if needed), acetaminophen and non-steroidal anti-inflammatory drugs; caffeine and short-term opioids may be offered. In order to maximize availability and uptake of these potentially mitigating strategies, proactive follow-up particularly of patients at highest risk is crucial.

WHAT ARE LONG-TERM COMPLICATIONS AFTER NEURAXIAL PROCEDURES?

The potential for long-term complications is increasingly recognized. These include serious sequelae of chronic pain, depression, subdural hematoma and cerebral venous sinus thrombosis, particularly described among obstetric patients. It remains unclear whether EBP mitigates, prevents, treats or exacerbates these sequelae. Severe complications such as cerebral sinus vein thrombosis and subdural hematoma may be missed in the setting of PDPH.

WHERE DO WE GO FROM HERE?

These multisocietal guidelines about DP, intentional and unintentional, are a first step in the right direction to raise awareness surrounding the importance of (1) adequate consent and education, (2) use of appropriate needles and technique, (3) prevention and management of PDPH (mostly EBP), and (4) appropriate follow-up and monitoring of worsening symptoms.

Cutting needles are routinely used for intentional DP, inexperienced clinicians perform DP or labor epidural analgesia with insufficient training and supervision, and lack of proactive follow-up systems, continue to contribute to the burden of PDPH. These guidelines should serve as a call to action to focus on modifiable factors to minimize the occurrence of DP, and to report and treat long-term sequelae.

Some gaps in evidence are noted in these guidelines, in particular the fact that none of the many proposed prophylactic measures in the setting of unintended DP (eg, placement of an intrathecal catheter or prophylactic blood patch via an existing epidural catheter to prevent PDPH), nor conservative measures with both intended or unintended DP (eg, hydrocortisone, theophylline or gabapentin) or procedural measures (sphenopalatine ganglion block, epidural dextran or gelatin, or neuraxial morphine) showed sufficient benefits to be recommended routinely. This suggests the need to improve the evidence surrounding the circumstances of a DP, whether intended or unintended, and patient characteristics for which such measures may be beneficial. Absence of evidence does not mean evidence of absence.

The next step, not addressed in these guidelines, will be to reassess these broad recommendations with a context-specific (diagnostic vs analgesic/anesthetic procedure, ambulatory vs in-patient), patient-specific (obstetric, adult, pediatric, and individual risk factors) and procedure-specific (unintended vs intended DP, type of needle) perspective. Risk factors and preventative measures will need to be contextualized, bearing in mind that the obstetric population is at high risk for chronic issues, including postpartum depression, which were not addressed here.

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REFERENCES

- 1 Mims SC, Tan HS, Sun K, *et al.* Long-term morbidities following unintentional dural puncture in obstetric patients: a systematic review and meta-analysis. *J Clin Anesth* 2022;79:110787.
- 2 Guglielminotti J, Landau R, Li G. Major neurologic complications associated with postdural puncture headache in obstetrics: a retrospective cohort study. *Anesth Analg* 2019;129:1328–36.
- 3 Moore AR, Wieczorek PM, Carvalho JCA. Association between post-dural puncture headache after neuraxial anesthesia in childbirth and intracranial subdural hematoma. *JAMA Neurol* 2020;77:65–72.
- 4 Uppal V, Russell R, Sondekoppam R, *et al.* Consensus practice guidelines on postdural puncture headache from a multisociety, international working group: a summary report. *JAMA Netw Open* 2023;6:e2325387.
- 5 Uppal V, Russell R, Sondekoppam RV, *et al.* Evidence-based clinical practice guidelines on postdural puncture headache: a consensus report from a multisociety international working group. *Reg Anesth Pain Med* 2023:rapm-2023-104817.
- 6 Rochweg B, Almenawer SA, Siemieniuk RAC, *et al.* Atraumatic (pencil-point) versus conventional needles for lumbar puncture: a clinical practice guideline. *BMJ* 2018;361:k1920.

